

Exploring Integrated and Decision-Weighted Gradients for Image Attribution

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Abstract—This paper describes two from-scratch PyTorch attribution notebooks for a pretrained ImageNet ResNet-50: Integrated Gradients (IG) and an Integrated Decision Gradients (IDG)-inspired, decision-weighted variant. Both trace a straight path from a zero tensor to a normalized image and use model gradients to assign pixel importance. The second notebook allocates samples according to changes in the target logit and additionally weights gradients by the directional derivative along the path. The repository presents heatmaps for a macaque example, with the decision-weighted output appearing more concentrated on the animal than the IG output. This is a qualitative observation only. The notebooks use different example images by default, and no quantitative faithfulness or stability benchmark is reported; consequently, the displayed examples cannot establish comparative superiority.

Index Terms—explainable AI, integrated gradients, image attribution, ResNet-50, gradient saturation

I. Introduction

Attribution methods assign scores to input pixels to help examine why an image classifier selected a class. A heatmap can support model inspection, but its visual sharpness is not itself proof that the explanation is faithful. Integrated Gradients (IG) integrates the model’s input gradient along a path from a reference input to the image and was introduced with axiomatic motivation [1]. Integrated Decision Gradients (IDG) addresses path regions where the output changes little, known as saturation, by concentrating attention near consequential score changes [2]. This project implements IG and a decision-weighted variant in separate PyTorch notebooks without attribution libraries [3].

The aim here is to document the actual code and its visible outputs, then distinguish the observed heatmaps from claims that would require controlled evaluation. This distinction is especially relevant because both baseline selection and numerical integration can affect an attribution map.

II. Common Experimental Setup

Both notebooks load the torchvision ResNet-50 weights named IMAGENET1K_V1, set the model to evaluation mode, resize an image to 256 pixels, take a 224×224 center crop, convert it to a tensor, and apply ImageNet channel normalization. The predicted top class becomes the attribution target. The baseline is a zero tensor in the normalized input space. Thus “zero baseline” here means zero after normalization; it should not automatically be interpreted as a black RGB image.

Let $F_c(x)$ be the selected class logit, x' the baseline, x the normalized input, and $\Delta = x - x'$. The shared path is $\gamma(\alpha) = x' + \alpha\Delta$ for $0 \leq \alpha \leq 1$. Attribution is computed for the target logit rather than a softmax probability. For visualization, the notebooks sum the absolute attribution over the three color channels at each pixel and display the result as a heatmap. Absolute-value aggregation removes sign, so the plots do not distinguish evidence supporting the class from evidence opposing it.

III. Integrated Gradients Implementation

For input coordinate i , continuous IG is

$$\text{IG}_i(x) = \Delta_i \int_0^1 \frac{\partial F_c(\gamma(\alpha))}{\partial x_i} d\alpha. \quad (1)$$

The notebook samples 50 uniformly spaced path positions, including both endpoints, obtains the input gradient at each position by backpropagation, averages these gradients, and multiplies elementwise by Δ [3]. This is a simple numerical approximation of the integral. Because it averages endpoint samples without trapezoidal endpoint weights, it is not the trapezoidal rule; numerical completeness should be checked if quantitative use is intended. The implementation uses a single default test image of a dog, although another image can be substituted.

IV. Decision-Weighted Variant

The second notebook first partitions the path into 50 coarse intervals. For interval j , it computes the target-logit change $F_c(\gamma((j+1)/50)) - F_c(\gamma(j/50))$ and rounds that interval’s fraction of the total endpoint logit change into a requested number of samples. Intervals assigned zero or negative samples are skipped. At each retained point it computes a gradient $g = \nabla_x F_c(\gamma(\alpha))$ and a directional-derivative weight $w = g \cdot \Delta$. This gives greater nominal emphasis to parts of the path where the class logit changes rapidly.

The code accumulates terms proportional to $g_i w$, scaled by each interval’s sample count, multiplies by Δ_i , and performs a final normalization by the number of retained steps [3]. This is an IDG-inspired implementation, but the additional normalization and rounded, potentially skipped intervals mean its numerical behavior should be verified against the IDG paper’s specification [2]. When the endpoint logit change is near zero, the allocation ratio can also become unstable. Neither convergence tests across step counts nor a completeness check appear in the repository.

V. Qualitative Result and Discussion

The repository includes a preprocessed macaque image and heatmaps described as IG and IDG results [3]. In the displayed example, the IG visualization contains fine texture and peripheral activity, while the decision-weighted visualization appears more focused around the animal’s face and body. That observation describes the published images, not a measured increase in explanation quality. The two notebooks default to different inputs (dog for IG and monkey for the decision-weighted variant), so a controlled side-by-side rerun on the same image and target class is needed before treating their outputs as a method comparison.

A stronger experiment would process the same ImageNet validation images, classes, normalization, baseline, and color scale with both methods. Insertion and deletion curves, sensitivity to baseline and step count, runtime, and completeness error would then provide quantitative context. Reporting uncertainty across images would help distinguish a repeatable effect from a compelling single example. Heatmap concentration alone could even hide

relevant but spatially distributed evidence, so it should not be the sole criterion.

VI. Conclusion

The notebooks provide readable implementations of path-based attribution for ResNet-50 and illustrate how decision-sensitive weighting changes a displayed heatmap. The evidence remains exploratory: the example is qualitative, defaults differ across notebooks, and the decision-weighted estimator needs numerical validation. The next step is a controlled same-image evaluation with explicit faithfulness metrics.

References

- [1] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic attribution for deep networks,” in Proc. 34th Int. Conf. Machine Learning, 2017, pp. 3319–3328.
- [2] C. Walker, S. K. Jha, K. Chen, and R. Ewetz, “Integrated decision gradients: Compute your attributions where the model makes its decision,” in Proc. AAAI Conf. Artificial Intelligence, 2024.
- [3] T. Akolawala, “Gradient-Based Attributions,” GitHub repository, 2026. [Online]. Available: github.com/taherakolawala/gradient-based-attributions